

BP³: Improving Cuff-less Blood Pressure Monitoring Performance by Fusing mmWave Pulse Wave Sensing and Physiological Factors

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Abstract

Cuff-less methods, especially pulse wave analysis (PWA) techniques with PPG/mmWave sensing, have shown great potential for non-intrusive blood pressure (BP) monitoring. However, the state-of-the-art solutions are only validated on small-scale healthy subjects, neglecting patients with abnormal BP and thus a more urgent need for BP monitoring. To bridge the gap, we first build the largest mmWave-BP dataset to our knowledge, including 930 real patients with cardiovascular diseases, and perform extensive experiments, which reveals that all existing PWA methods exhibit far less satisfactory performance with standard deviation errors (STD) exceeding 16 mmHg for systolic BP (SBP) and 11mmHg for diastolic BP (DBP). An in-depth investigation shows that physiological factors have complex effect on vascular elasticity and structure, thus people with very different BP values may exhibit extremely similar pulse waveform, which leads to confusion in model learning. In this work, we propose BP³, which fuses physiological factors into sensing-data-driven deep-learning framework, so as to capture the intricate effect of physiological factors during the whole process of learning pulse waveforms. Evaluation results show that BP³ achieves the mean errors of -1.57 mmHg and -0.34 mmHg, STD of 9.77 mmHg and 7.93 mmHg for SBP and DBP, respectively. Moreover importantly, BP³ shows remarkable gain particularly for subjects with abnormal BP, achieving mean errors that are only 0.48% ~ 20.86% of the state-of-the-art solutions.

CCS Concepts

• Human-centered computing → Ubiquitous and mobile computing design and evaluation methods; • Applied computing → Health informatics.

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1 Introduction

As of 2023, an estimated 1.28 billion adults worldwide globally were suffering from hypertension [18]. Hypertension notably elevates the risk of severe complications, including stroke, coronary artery disease, heart failure, and end-stage renal disease. These conditions account for 13.5% of all deaths worldwide each year, leading hypertension as the foremost risk factor for cardiovascular diseases [24]. Despite its prevalence and severe implications, approximately 580 million individuals with hypertension remain undiagnosed. In countries such as the United Kingdom and China, the awareness rates are alarmingly low, at 58.4% and 56.4% respectively [7]. This lack of awareness underscores the critical need to improve the accessibility and comfort of blood pressure (BP) monitoring devices.

The predominant technique currently used for measuring BP is cuff device [33], which needs an inflatable cuff to incrementally pressurize the user's arm. This process can be notably uncomfortable, especially for those with sensitive skin or elevated blood pressure. In light of these drawbacks, there is a growing interest in developing cuff-less measurement methods that offer more comfortable and convenient alternatives, including pulse transit time (PTT) based methods [10, 13, 21], and pulse wave analysis (PWA) methods [4, 26, 35, 36, 43], by leveraging sensing modalities such as PPG, mmWave and ultrasound, etc.

While showing great potential, we find that existing works are mainly verified over small-scale datasets consisting of healthy subjects [4, 5, 26, 27, 54], and their performance is far less satisfactory when confronted with a larger scale population, particularly for patients with abnormal BP. In particular, we build the largest mmWave sensing based BP dataset with 1,012 subjects, among them 930 are real patients with cardiovascular diseases and real-world

BP range. Using the dataset, we evaluate three state-of-the-art algorithms [4, 26, 46], results show that the standard deviation error (STD) of SBP and DBP exceed 17 mmHg and 11 mmHg, much worse than the performance on previous smaller datasets, (e.g., 4.8 mmHg for SBP and 3.79 mmHg for DBP reported in [26]). An in-depth analysis shows the underlying reason, *i.e.*, previous works focus on distilling BP features solely from pulse waveform, yet neglect the impact of physiological factors. For instance, the waveforms of two subjects in our dataset are extremely similar, with a Pearson correlation coefficient of 0.99, yet one is a 78-year-old male with Grade 2 hypertension, while the other is a 59-year-old female with normal blood pressure, and their BP difference exceeds 50 mmHg. We denote this as a multivalued mapping problem.

In this work, we propose BP^3 , a cuff-less Blood Pressure measurement system by fusing mmWave Pulse wave sensing and Physiological factors to enhance the accuracy of BP estimation, particularly for users with abnormal BP values. We illustrate the usage scenarios of BP^3 in Figure 1, where BP^3 captures a subject's pulse waves using a wearable prototype (*i.e.*, a smartwatch) equipped with a COTS mmWave radar, and then feeds the waveform, along with the subject's physiological factors into a custom-built deep-learning framework, which fusing the information and output subject's BP values.

Despite of conceptual simplicity, the major challenge in realizing BP^3 lies in that, *how to fuse the two different modalities, i.e., the pulse wave time series and the static physiological factors, so as to faithfully capture the impacting effect of physiological factors?* Most existing research on BP prediction does not consider the impact of physiological information [31, 42, 46]. While a few studies made attempts to include these factors, their methods are generally in a straightforward manner such as direct concatenation [26] or weighted sum [4] techniques. However, these approaches fall short in fully leveraging their intricate interactions. Physiological factors such as age and BMI alter vascular elasticity and structure [37], which then exert varying influences on different parts of the pulse waveform. To tackle the challenge, we introduce a multimodal fusion learning framework, which consists of two modules: (i) a multimodal embedding module to separately process each modality, in which we adapt patch-wise embedding and rotary position embedding mechanism to represent the temporary pulse waveforms and MLP blocks to process the discrete physiological factors. (ii) a physio-pulse attention module, which enables physiological factors to dynamically guide the learning of pulse waveforms. It is noteworthy that, although our physio-pulse attention module is built upon the widely-used attention mechanism in deep learning, we have specifically tailored it for capturing the complex interactions between modalities. In particular, we integrate a bi-attention mechanism that not only learns the internal relationships among different pulse waveform patches, but also captures the interactions between the pulse waveform and individualized physiological factors. As a result, these modules enhances the ability of BP^3 to achieve accurate BP estimation.

We use 5-fold cross validation to validate the performance of our proposed method. Overall, we can achieve accurate estimates of SBP and DBP with the mean error (ME) of -1.57 and -0.34 mmHg, the STD of 9.77 and 7.93 mmHg on our large-scale dataset, where DBP meets the FDA's AAMI protocol [47] (*i.e.* ME ≤ 5 mmHg, STD ≤ 8 mmHg), while SBP is close to it. It is noteworthy that our method

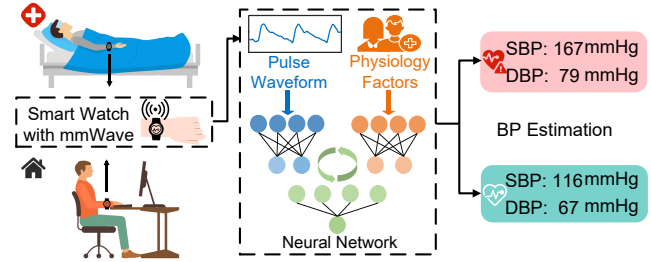


Figure 1: Illustration of BP^3 .

outperforms the state-of-the-art PWA methods and demonstrates exceptional robustness, particularly for subjects with abnormal BP, *e.g.* a reduction of more than 83% on ME and 44% on the mean absolute error (MAE) for patients with Grade 2 hypertension. Additionally, the visualization results confirm that our method effectively highlights different parts of the waveform individually, enhancing its interpretability.

Our contributions are summarized as follows:

- Through large-scale experiments over one thousand subjects, we have identified a general problem in applying PWA to complex real-world measurements. Our findings show the importance of considering physiological factors in pulse waveform analysis based methods.
- We design a novel multimodal fusion frame for temporal pulse waves and static personal information by cross-attention mechanism, so as to solve the multivalued mapping problem and implement a more universal cuff-less BP measurement method.
- To our knowledge, we have built the largest mmWave-BP dataset for cardiovascular patients in collaboration with professional medical institution, which will be released to support future study.

2 Measurement

In this section, we first introduce our dataset in Section 2.1 and use it to evaluate existing methods in Section 2.2, during which we will uncover unexpected phenomena and investigate the impact of physiological factors on pulse waveforms in Section 2.3.

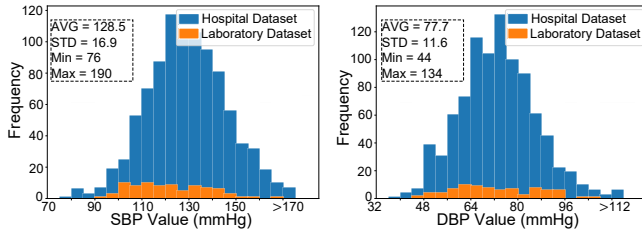
2.1 Dataset Description

Over the eight-month period, we recruited a total of 1,012 subjects and used our mmWave sensing device to build the largest mmWave-BP dataset to our knowledge. Detailed information about the data collection process will be discussed in Section 4.1. Compared to existing work [9, 11, 17, 44], our mmWave-BP dataset is distinguished by its large-scale and realistic patient data.

(i) *Large-scale population with diversity.* 930 subjects are hospitalized patients in Cardiology and Cardiovascular Surgery Department of hospital, who inherently have a heightened concern for their BP and indeed require daily BP monitoring in the hospital. The remaining 82 subjects are healthy individuals collected from our laboratory. Overall, the dataset comprises 2,035 records and over 71,225 pulse cycles. It is noteworthy that our dataset exhibits a significantly

Table 1: Demographics of hospital subjects.

Gender	female	372(36.8 %)	Gender	male	640 (63.2%)
Age	<45 (young)	182 (18.0%)	BMI (kg/m ²)	<18.5 (underweight)	63 (6.2%)
	45-60 (middle)	223 (22.0%)		18.5-25 (healthy)	492 (48.6%)
	60-90 (elderly)	588 (58.1%)		25-30 (overweight)	352 (34.8%)
	>90 (extremely old)	19 (1.9%)		>30 (obesity)	195 (10.4%)

**Figure 2: Distribution of BP.**

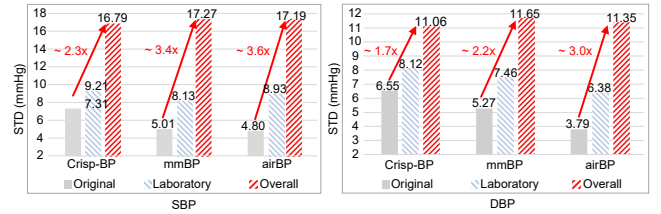
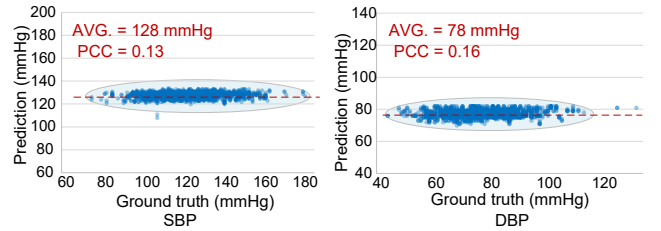
greater representation of individuals with high risk physiological factors compared to the general population. As demonstrated in Table 1, 58.1% population is elderly individuals aged 60 to 90 [3], and the prevalence of overweight and obese individuals is 45.2%, as evidenced by the BMI measurements [29]. The rich diversity presents challenges for developing robust and universal BP monitoring methods.

(ii) *Wide BP range.* In Figure 2, we present bar charts to show the BP distribution of subjects from hospital and laboratory, respectively. Specifically, the hospital readings extend from 76 to 190 mmHg for SBP, and 44 to 134 mmHg for DBP, which includes 27.05% of subjects in Grade 1 hypertension (140-159 mmHg) and 6.22% in Grade 2 hypertension (≥ 160 mmHg), both higher than that in general public [38] on the basis of the 2020 ISH Global Hypertension Practice Guidelines [51]. On the other hand, the laboratory part is more concentrated within a smaller range and the majority of subjects have BP within the normal level.

In addition to the diversity and high-risk nature of the subjects, our dataset is collected in the real-world environment of a hospital. The complexity of the hospital environment may introduce additional noise and interference from various sources, such as other medical devices, personnel activities, or the movements of patients themselves. These factors can lead to degradation of the mmWave signals, necessitating further processing to ensure data quality and reliability.

2.2 Evaluation of Existing Methods

We evaluate three state-of-the-art cuff-less BP measurement methods, *i.e.*, crispBP [4], mmBP [46] and airBP [26] on our dataset (details of these methods are given in Section 4.3). We compare these methods on three datasets: the dataset used in the original paper, our laboratory dataset, and our overall dataset, and depict the results in Figure 3. We have two major observations: (i) According to the statistics reported in their original paper, the three methods all meet the FDA's AAMI protocol with low error. Results over our

**Figure 3: The accuracy of the state-of-the-art across original, laboratory and overall datasets.****Figure 4: Correlation diagram of airBP on our dataset.**

laboratory dataset of 82 subjects corroborate with this, which show that there are slight decreases in accuracy, with STD increased by only about 2 mmHg. (ii) However, when evaluated on the entire dataset comprising a thousand subjects, the errors increase significantly. Specifically, the STD reaches up to 17mmHg for SBP, and 11mmHg for DBP, which are 1.7 ~ 3.6 times the original.

Moreover, we have a microscope investigation as shown in Figure 4, where we compare the measurement output of airBP against the ground-truth. Clearly, the outputs consistently hover around a specific value, with SBP averaging 128 mmHg and DBP averaging 78 mmHg, which happens to be the exact average value of the ground-truth BP of our dataset. In addition, while the ground-truth values exhibit a wide distribution, with SBP spanning 114 mmHg, the measurement values show less than 25% of the variation. Both Crisp-BP and mmBP exhibit similar patterns, implying that relying solely on pulse waveform is insufficient.

2.3 Understanding the Impact of Physiological Factors

As noted before, existing cuff-less methods particularly the PWA-based methods, infer BP values by extracting pulse waveform features in terms of amplitude, temporal characteristics, and morphology. This implies that they rely on the specificity of individual pulse waveforms. For instance in Figure 5(a), the two subjects exhibit quite different pulse waveform, and lead to different BP values. However, there exists opposite cases. As shown in Figure 5(b), two subjects with a large BP discrepancy (*i.e.*, 51mmHg in SBP) can exhibit nearly identical pulse waveforms. We define such issues as multivalued mapping, *i.e.*, waveforms with extreme similarity are associated with significantly different BP values.

We find that multivalued mapping is not a corner case, but occurs frequently. To be precise, we cut each waveform into 256-length segments and pair the segments from different subjects. Then we

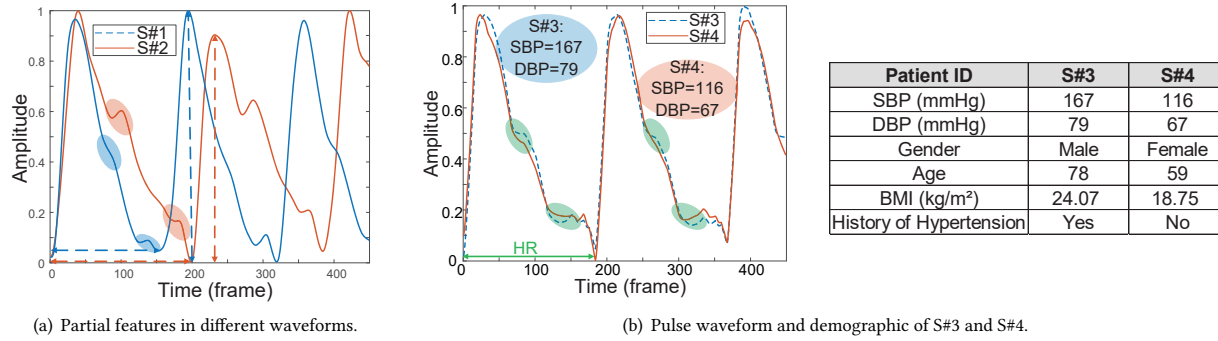


Figure 5: PWA methods and example of multivalued mapping problem .

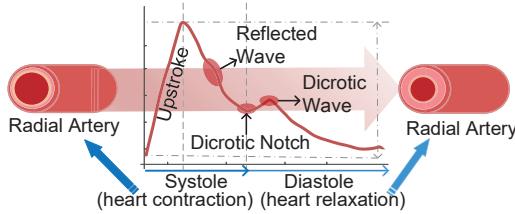


Figure 6: The artery diameter variance and corresponding pulse waveform measured by mmWave.

use the dynamic time warping (DTW) algorithm and quantify their similarity by calculating the Pearson correlation coefficient. When the correlation coefficient of a pair of waveforms exceeds 0.90 and the corresponding BP values differ by more than 8 mmHg, we consider this pair of waveforms to exhibit a multivalued mapping relationship. In this way, approximately 13.2% pulse waveforms in our dataset exhibit multivalued mapping, originating from 9.5% of the subjects. Furthermore, this phenomenon is not unique to our dataset, but also occurs in other dataset, e.g., 16.2% in the Blumio dataset [11], a mmWave dataset including 105 subjects, and 15.0% in MIMIC-III [17], a widely applied dataset including perceived data obtained by PPG. This widespread occurrence across different datasets highlights the limitations of current PWA methods, as there may not exist one-to-one mapping between pulse waveforms and BP values. The problem underscores the need for a improved approaches to enhance the understanding of the relationship between them.

The formulation of pulse waveform. In essence, the pulse waveform results from changes in the diameter of arterial blood vessel during its dilation and relaxation. In Figure 6, we demonstrate the pulse waveform generated as the dynamic change of artery diameter throughout a cardiac cycle. In particular, as the contraction and relaxation of blood vessels, the waveform shows upstroke and downward trends. Moreover, the branches of the arterial network and the activity of the heart valves can also create other small waves. During this process of vascular change, various individual factors including non-modifiable ones and modifiable cardiovascular risk factors can affect the elasticity and structure of blood vessels. Consequently, these influences are reflected in the structure and elasticity of blood vessels, affecting different stages of the pulse wave to varying degrees [14].

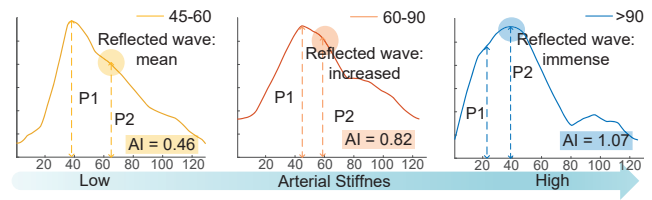


Figure 7: Diagram of the impacts of age on pulse waveforms.

Here, we focus on four factors: **age, gender, obesity and history of abnormal BP**. Existing research has proved that these physiological factors significantly impact cardiovascular health and are relatively easy to obtain:

(i) **Age:** As individuals age, their arterial stiffness gradually increase. This leads to reduced vasodilation during the ejection phase and an earlier return of reflected waves [19]. The Augmentation Index (AIx), which is calculated by the ratio of the amplitude of the ejection period $P1$ to the amplitude of the reflected wave $P2$, increases with age, even exceeding 1 [34]. Figure 7 shows the variations of waveform as age increases. (ii) **Gender:** Compared with females, the pulse waves in man are usually deeper and slower, with a longer diastolic proportion and a shorter systolic proportion while the amplitude of the pulse wave is higher [12]. (iii) **Obesity:** There are often notable changes in dermal thickness and radial artery depth as obesity severity increases [22], which may contribute to the complexity and reliability of measurement. (iv) **Hypertension:** In hypertensive patients, the peak value of the main wave is typically higher, reflecting increased cardiac load and arterial wall pressure. It often leads to reduced arterial elasticity, which causes reflected waves to return to the heart earlier [23, 41].

The influence of varying effects and degrees of physiological factors can lead to diverse pulse waveform patterns. Figure 5(b) also demonstrates the corresponding physiological factors of S#3 and S#4. Despite their pulse waves exhibiting remarkable overlap, there is a 51 mmHg difference in SBP and a 12 mmHg difference in DBP between them. Besides, S#3 is a 79-year-old male with Grade 2 Hypertension and S#4 is a 59-year-old female with normal BP value. This suggests that the combination of these physiological discrepancies can lead to similarities in waveform shapes. **Therefore, while inferring BP solely based on pulse waves can be unreliable, it is crucial to consider various physiological factors to improve accuracy.**

3 Design of BP³

3.1 System Overview

In Figure 8, we present the overview of BP³, which consists of three key modules: Data Preprocessing, Multimodal Embedding and Physio-Pulse Attention. The first module focuses on collecting and processing mmWave data, and the latter two are in charge of estimating BP. They are designed to sense subtle variations of pulse waveforms and model the complex relation with physiological factors. We will first provide an overall summary of the framework, followed by a detailed description of each module and its function.

Data Preprocessing. We utilize a wearable mmWave device to transmit and receive signals for capturing pulse information from the wrists. From thousands of subjects in hospital and laboratory settings, we have constructed a mmWave-BP dataset. The collection procedure and processing methods are described in details in Section 4.

Multimodal Embedding. We process temporal data and discrete data separately for effectively integrating multimodal data. To capture the tiny movements of radial artery, we use multi-antenna mmWave signals as input and customized design a network to extract local feature map of pulse signals. Existing waveform-based methods typically treat single or multiple heartbeats as whole inputs [26, 42, 55], which may obscure subtle variations and intricate patterns within the waveform. In contrast, by dividing the feature map into patches, we can effectively capture subtle changes and internal patterns of the pulse waveform. Considering the periodicity and fixed pattern of pulse waveform, we introduce rotary position embedding [48] (RoPE), which is a mechanism that can model the relative positions between each patch in the waveform. For discrete physiological index, we use MLP blocks to process it.

Physio-pulse Attention. In order to accurately model the complex interactions between modalities, we introduce the physio-pulse attention module, a bi-attention mechanism designed to guide waveform learning based on individual physiological factors. By combining the self-attention and cross-attention mechanism for different modalities, this module assigns dynamic weights to different patches of pulse waves based on the unique physiological information of each individual. It not only tackles the problem of multivalued mapping but also enhances interpretability by visualizing the attention weights, providing insights into how different physiological factors influence the learning process.

3.2 Multimodal Embedding

To address the multivalued mapping problem between BP and pulse waveforms, BP³ focus on the fusion of pulse waveforms and physiological factors. Given the fundamentally different structures between the temporal data and discrete indicator, we design separate embeddings procession for each modality.

Pulse waveform. To capture the subtle change of blood vessel diameter with an amplitude of around $25 \pm 5 \mu\text{m}$ [20], we collect pulse waves from the received signal strength (RSS) of mmWave signals of multi antennas pairs. Due to the variations of antenna position, this can provide richer information of pulse waves. As shown in Figure 9, we employ RSS of each antenna pair as a separate channel and then design a feature extraction layer to capture the information within waveforms, which consists of one convolutional

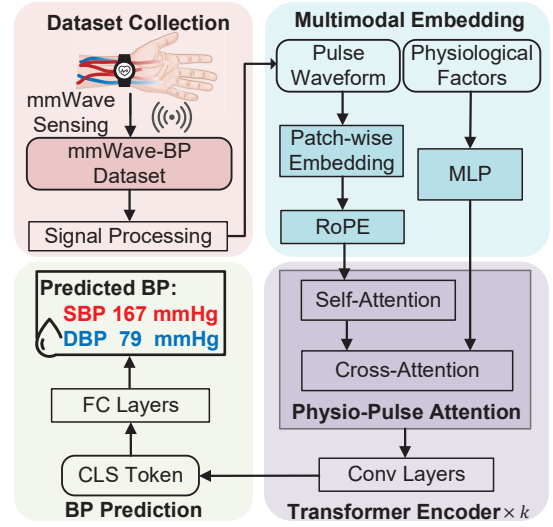


Figure 8: System overview of BP³.

layer followed by four ResBlocks. The convolution layer is crucial for extracting temporal pulse details, which is demonstrated as (feature map count, kernel size). To facilitate deep pulse feature learning, the 4 ResBlocks are grouped into 2 stages of equal size. The count of feature map $C = 4$ is multiplied by 2 at each stage for deeper pulse waveform feature extraction. Residual connections are incorporated in each ResBlock to improve gradient during back-propagation.

Then, to effectively analyze both local and global information within the feature map, we segment it into N patches. A single-layer MLP is applied to generate a fixed-size embedding vector, which projects dependencies across channels and adjacent patches in temporal order. Additionally, we introduce a learnable class token (CLS token) at the beginning of the patches, which will aggregate global information across the patches. The embedding of pulse waveform can be formulated as $X_{pulse} \in \mathbb{R}^{(N+1) \times d}$.

Although various waveforms across different subjects encompass different details, they basically shaped with a systolic peak and bulges as Figure 6 shows. Therefore, we introduce RoPE to better understand the periodic characteristics and prepare for the subsequent attention mechanism. Particularly, RoPE employs a rotation matrix to encode absolute positions while simultaneously integrating explicit relative position dependencies. Given two samples of waveform patch tokens x_m, x_n , whose absolute positions are m and n , the RoPE function $f_q(\cdot), f_k(\cdot)$ applicable to queries and keys can be expressed as:

$$\begin{aligned} f_q(x_m, m) &= R_{\Theta, m}^d x_m W_q \\ f_k(x_n, n) &= R_{\Theta, n}^d x_n W_k \end{aligned} \quad (1)$$

$$R_{\theta, m} = \begin{pmatrix} \alpha_1 & & & \\ & \alpha_2 & & \\ & & \ddots & \\ & & & \alpha_{d/2} \end{pmatrix}, \alpha_i = \begin{pmatrix} \cos(m\theta) & -\sin(m\theta) \\ \sin(m\theta) & \cos(m\theta) \end{pmatrix} \quad (2)$$

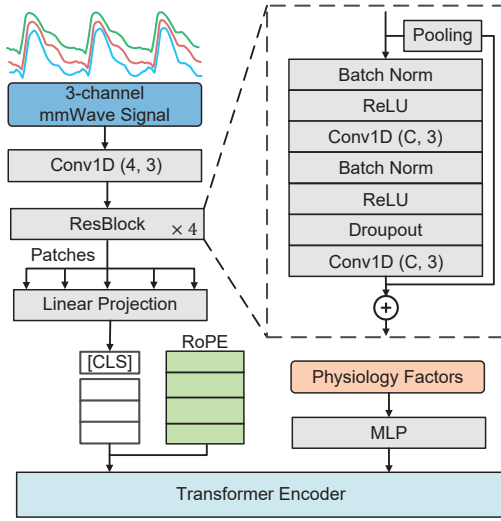


Figure 9: Architecture for multimodal embedding generation.

where d represents the dimension, $R_{\Theta, m}^d$ represents rotary matrix, $\Theta = \{\theta_i = 1000^{-2(i-1)/d}, i \in [1, 2, \dots, d/2]\}$ is the set of rotary angle θ and W_q, W_k are the linear transformation matrix for query and key vectors respectively. Then, the multiplication of $R_{\Theta, m}^d$ and $x_m = q_1, q_2, \dots, q_d$ is realized to calculate the rotation matrix multiplication:

$$R_{\Theta, m}^d x_m = \begin{pmatrix} q_1 \\ q_2 \\ \dots \\ q_{d-1} \\ q_d \end{pmatrix} \otimes \begin{pmatrix} \cos m\theta_1 \\ \cos m\theta_1 \\ \dots \\ \cos m\theta_{d/2} \\ \cos m\theta_{d/2} \end{pmatrix} + \begin{pmatrix} -q_2 \\ q_1 \\ \dots \\ -q_d \\ q_d \end{pmatrix} \otimes \begin{pmatrix} \sin m\theta_1 \\ \sin m\theta_1 \\ \dots \\ \sin \theta_{d/2} \\ \sin \theta_{d/2} \end{pmatrix} \quad (3)$$

In this way, absolute positional information is added to the embedding through rotational transformation.

Physiological factors. We consider 6 factors as physiological factors. Specifically, we treat gender and history of abnormal BP as discrete values. The history of abnormal BP is categorized into three groups: individuals with a history of hypertension, those with hypotension, and those with no prior abnormalities. Age, height, weight, and BMI are considered as continuous values and are normalized respectively. Hence, we derived a 6-dimensional vector [gender, age, height, weight, BMI, BP_history] for each subject. In order to proceed with the subsequent steps, we use MLP block to align it with the dimensions of waveform embedding, resulting the feature map $X_{Physio} \in \mathbb{R}^{1 \times d}$. Then, physiological factors PF along with pulse patches tokens are fed into the customized transformer encoder.

3.3 Physio-Pulse Attention

After obtaining the embeddings of pulse waveforms X_{Pulse} and physiological factors X_{Physio} , few studies attempt to calibrate by combining the two using a straight-forward manner such as weighted sum or concatenation [4, 26]. However, all of these PWA methods overlook the interaction between the two. As mentioned in Section 2.3, personal physiological factors modulate the shape of the

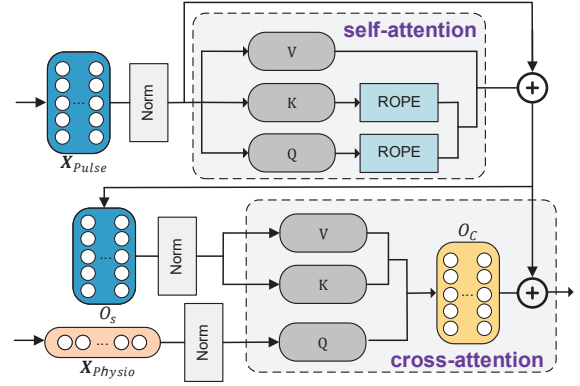


Figure 10: Physio-pulse attention mechanism.

pulse waveform, with each factor contributing differently to the overall waveform at various stages, rather than acting as isolated features. Hence, we design an encoder consists of two key modules: the physio-pulse attention as shown in Figure 10 for fusing the embeddings of different modalities, and the convolutional layer for further enhancing local features.

Physio-pulse attention. On the basis of rotary position embedding, we use self-attention mechanism to accurately utilize positional relationships in pulse waveform within patches. The attention score, computed as the inner product of the query at position m with the key at position n of the pulse waveform embedding, can be expressed as:

$$\begin{aligned} Score_{m,n} &= (R_{\Theta, m}^d x_m W_q)^T (R_{\Theta, n}^d x_n W_k) \\ &= x^T W_q R_{\Theta, n-m}^d W_k x_n \end{aligned} \quad (4)$$

This attention score can reflect the relative positional $n - m$ of the input sequence and the attention output can be obtained by:

$$Attn_m = softmax(\frac{\sum_{n=1}^N Score_{m,n} \cdot V_n}{\sqrt{d}}) \quad (5)$$

where $m \in [1, 2, \dots, N]$ and these attention output constitute the final output of self-attention mechanism $O_s = [Attn_1, Attn_2, \dots, Attn_N]$. In this way, the long-distance dependency relationships and their relative fixed waveform structure between patches can be learned.

To individually fuse the features of physiological factors and pulse waveforms, we then design the cross-attention mechanism. Specifically, to align with the waveform, we broadcast the dimension of the embeddings of physiological factors, and then serve it as the query to actively seek relevant information within the waveform features while serve the output of self-attention as the key and value. The query Q , key K , value V and the output of the cross-attention O_c can be defined as follows:

$$\begin{aligned} Q &= X_{Physio} W_Q, \quad K = O_s W_K, \quad V = O_s W_V \\ O_c &= softmax(\frac{QK^T}{\sqrt{d}})V \end{aligned} \quad (6)$$

where W is corresponding learnable weight parameters for transformation. This structure enables the model to dynamically adjust the attention for each patch token, according to the current physiological context provided by the query. Additionally, we use residual

Algorithm 1 Algorithm of our transformer encoder.

Input: A bag of feature embeddings $X_{Pulse} \in \mathbb{R}^{(N+1) \times d}$, $X_{Physio} \in \mathbb{R}^{1 \times d}$.

Output: The feature representation $H^f \in \mathbb{R}^{(N+1) \times d}$ after fusion of physiological and pulse information.

- 1: Align dimensions
 $X_{Physio}^l \leftarrow \text{Broadcasting}(X_{Physio})$, where l denotes the layer index of the Transformer encoder, $X_{Physio}^l \in \mathbb{R}^{(N+1) \times d}$
- 2: Capture dependency between waveform patches
 $X_{Pulse}^l \leftarrow \text{SA}(X_{Pulse})$, where $X_{Pulse}^l \in \mathbb{R}^{(N+1) \times d}$
- 3: Fuse physiological factors and pulse waveforms
 $H \leftarrow \text{CA}(X_{Pulse}^l, X_{Physio}^l)$, where $H \in \mathbb{R}^{(N+1) \times d}$
- 4: Divide into patch tokens and class token
 $H_f, H_c \leftarrow \text{Split}(H)$, where $H_f \in \mathbb{R}^{N \times d}$, $H_c \in \mathbb{R}^{1 \times d}$
- 5: Convolutional layer
 $H_f^C \leftarrow \text{Conv}(H_f)$, where $H_f^C \in \mathbb{R}^{N \times d}$
- 6: Reconnect the class token
 $H^f \leftarrow \text{Concat}(H_f^C, H_c)$, $X_{Pulse} \leftarrow H^f$, where $H^f \in \mathbb{R}^{(N+1) \times d}$
- 7: Repeat step 2 to step 6 for k iterations

connection followed by both self-attention and cross-attention to maintain the flow of information.

Convolutional layer. After applying the two attention mechanisms that incorporate positional information, a convolutional layer is employed. The patch tokens are processed through layer normalization and a Conv1D layer to fuse spatial information while the class token is reconnected with them. The following steps either involve repeating the transformer encoder, where the attention mechanism aggregates the learned global features, or extracting the class token for the final estimation. Following k iterations of the transformer encoder, the class token can then be extracted for the final estimation. The pseudo-code for the encoder is shown in Algorithm 1.

3.4 BP Estimation

Finally, to perform the prediction task, we utilize the global representation derived from the class token and feed it into two MLP layers, each comprising two fully connected layers with ReLU as the activation function. The final output is the estimated SBP and DBP values. We train the learnable parameters in the proposed model with the MSE loss:

$$\text{Loss} = \frac{1}{N} \sum_{i=1}^N [(y_S - \bar{y}_S)^2 + (y_D - \bar{y}_D)^2] \quad (7)$$

where y_S, y_D represent the predicted SBP and DBP, $\bar{y}_S, \bar{y}_D$ represent the ground-truth values, respectively.

4 Implementation

In this section, we describe the details of the device and collection process in Section 4.1. The steps of preprocessing of data in Section 4.1, and the details of proposed model and training process in Section 4.2.

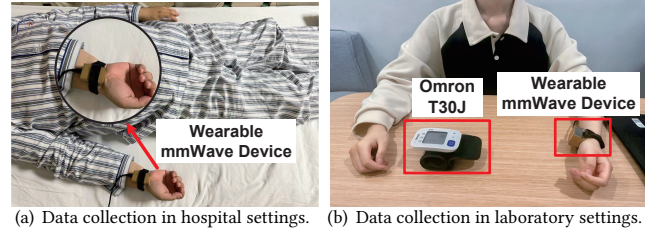


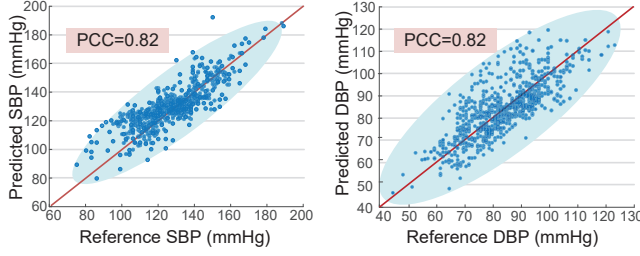
Figure 11: Data collection scenarios.

4.1 Data Collection and Preprocessing

Collection Process. Under the approval by the Institutional Review Board (IRB), we conducted measurements with each hospital subject in a supine position on the hospital bed as Figure 11(a) shows. Subjects were instructed to lay flat with both hands placed on either side of their body and legs stretched out, ensuring a standardized posture for data collection. Noted that patients exhibiting tendencies for upper limb infections or presenting wounds were systematically excluded from the study. For the laboratory subjects, as shown in Figure 11(b), each subject was required to sit in an adjustable-height chair to ensure that their wrist was aligned with heart level. The temperature in both the ward and laboratory was kept consistent, maintaining a stable and comfortable environment. For each data collection session, subjects were instructed to remain still and relaxed. Initial measurements of BP values were taken using an FDA-approved Omron wrist-cuff monitor device [1], serving as the ground-truth for our study. After a five-minute rest, our wearable mmWave device was placed on the same wrist to continuously collect data for approximately **35 seconds**, with two measurements taken for each subject to ensure data reliability.

Wearable mmWave prototype. Our prototype is a rectangular box equipped with an adjustable wristband, measuring 4.3 cm in length, 2.6 cm in width, and 2.5 cm in height. The box consists of five acrylic panels, with one side being cut out to ensure the transmission of the radar signal. Positioned at the bottom of the box, we have integrated a coin-sized COTS mmWave radar BGT60TR13C [2], along with an absorbing sponge. BGT60TR13C radar employs frequency-modulated continuous wave (FMCW) technology in the 60 GHz band. In our experiments, the frequency is from 58 GHz to 63.5 GHz, the range resolution is set to 2.73 cm and the ADC sample rate is 2 MHz. The number of samples per chirp is 32 and the sample rate for pulse wave sensing is 200 Hz, which is sufficient for pulse sensing. The power consumption of radar is only 270 mW, making it suitable for integration into mobile device such as smartwatches and smartphones. Although mmWave radar can achieve pulse monitoring from a longer distance [26], we place this compact box to adhere to the wrist without compressing blood vessels. This configuration better fits the shape of wearable watches and keeps the radar from contacting with the skin directly.

Data screening. A total of 1012 subjects were enrolled in the study, with 930 being recruited from hospital. During the BP measurement process using electronic sphygmomanometer as ground-truth, 14 subjects were found to have atrial fibrillation, which hindered the device's ability to obtain readings. 12 subjects were also

Figure 12: Correlation diagram of BP³.

excluded due to errors in recording BP values or physiological factors. Further, through data processing, 23 subjects were found to have poor pulse waveforms quality due to factors such as patient movement during the measurement, obesity causing deep arterial placement and other issues.

Data Preprocessing. All signal processing is executed using the MATLAB library. To extract the pulse information from the reflections of mmWave signals, we first perform range-FFT on the intermediate frequency (IF) signals to obtain data across various range bins. Given the relatively fixed position of the radar and wrist, we select the data from the second range bin and compute the RSS of the IF signals as input. To ensure the data quality, we drop the first and last 2.5 seconds of data. Then, we implement bandpass filtering within the frequency range of 0.4Hz to 7Hz to effectively filter out the potential influences of breathing. To reduce environmental noise, we employ the WDEN wavelet transform method on RSS. This automatic 1-D denoising function uses the 'sym4' wavelet as the basis and the decomposition level is 4. This method enables the decomposition of a signal into components at different scales, making it particularly suitable for non-stationary signals.

4.2 Details of proposed model and training

The learning model is built with Pytorch and TensorBoard libraries. In the multimodal embedding, we set the patch size $p = 16$, the number of patch $N = 375$, the embedding size $d = 96$ and the number of transformer blocks $k = 4$. In the output layer, we use two fully connected layers with $Linear(90, 32)$ and $Linear(32, 2)$. The training of neural networks leverages the formidable computational power of Tesla V100 GPU workstations to optimize performance and accuracy. We configure the batch size to 256 and train the model across 300 epochs using the SGD optimizer with an initial learning rate of $1e-4$. To ensure the robustness of our results, we implement 5-fold cross-validation at the subject level, ensuring that the same subject does not appear in both the training and testing sets simultaneously.

4.3 Comparison methods

In the measurement of Section 2.2 and evaluation in Section 5.1, we compare BP³ against three state-of-the-art methods.

- Crisp-BP [4] employs PPG signals to capture changes of blood volume and extracts BP-related features such as RWTT. It also combines [gender, age] with waveform features using a weighted

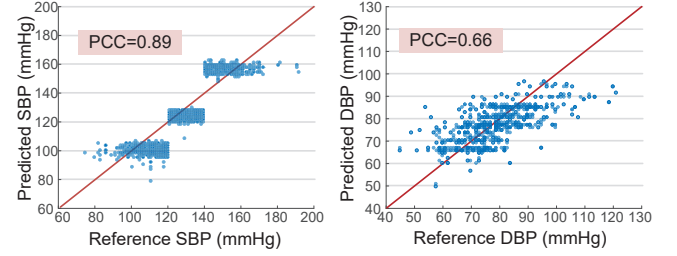


Figure 13: Correlation diagram of airBP [26].

sum to present a user-independent BP monitoring system. The result meet the FDA's AAMI protocol standards in a study involving 35 subjects.

- mmBP [46] separates noise from effective mmWave signals in the delay-Doppler domain and uses compensation signals to offset motion effects. By extracting 6 features and feeding them into regression models, mmBP achieves low error rates for 25 subjects.
- airBP [26] uncovers the correlation between RSS and arterial activity in mmWave technology and detects radial artery pulsations. By employing deep learning framework and concatenating 4 personal information with waveform features, it demonstrates high accuracy on 41 subjects.

5 Evaluation

In this section, we first evaluate the performance of BP³ against Crisp-BP [4], mmBP [46]. Then we validate the effects on populations with different physiological factors. We conduct micro-benchmark experiments to evaluate the effects of system modules and parameter configurations.

Evaluation metrics. To evaluate measurement performance, we use common metrics: mean error (ME), mean absolute error (MAE), standard deviation error (STD) and Pearson's Correlation (PCC). They can be defined as: $ME = \frac{\sum_{i=1}^I (\bar{x}_i - x_i)}{N}$, $MAE = \frac{\sum_{i=1}^I |\bar{x}_i - x_i|}{N}$, $STD = \sqrt{\frac{\sum_{i=1}^I (\bar{x}_i - x_i)^2}{N}}$, $PCC = \frac{\sum_{i=1}^I (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^I (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^I (y_i - \bar{y})^2}}$ where x is the SBP or DBP value predicted by methods and $\bar{x}$ is the ground truth BP value, I is the count of samples and i is the index.

5.1 Overall Performance.

Average accuracy over the whole dataset. By incorporating physiological factors consistent with our dataset, we compare the overall performance of BP³ with Crisp-BP [4] and airBP [26], both of which consider physiological influence, as shown in Table 2. From this, we observe: (i) ME and STD of BP³'s estimation are -1.57 and 9.77 mmHg for SBP, -0.34 and 7.93 mmHg for DBP, with DBP meets the standard requirements of AAMI and SBP results are close to the standards. (ii) In addition, we compare the PCC of SBP and DBP estimated by BP³ with the ground-truth, as shown in Figure 12. The x-axis represents the ground truth values, and the y-axis represents the estimated results. The points on the red line have identical values, and we observe that the data points closely adhere to this line. Specifically, BP³ achieves a high Pearson's correlation of 0.82 for both SBP and DBP. (iii) While comparison

Table 2: Performance comparison on overall dataset and abnormal BP dataset.

Dataset	Methods	Metrics (mmHg)		ME (SBP DBP)		STD↓ (SBP DBP)		MAE↓ (SBP DBP)		PCC↑ (SBP DBP)	
Overall Performance	CrispBP [4]	0.40	0.30	7.59	8.41	5.89	6.52	0.88	0.66		
	airBP [26]	0.21	-0.08	7.87	9.80	5.97	6.19	0.89	0.66		
	BP ³ (ours)	-1.57	-0.34	9.77	7.93	7.74	5.69	0.82	0.82		
Grade 2 hypertension	CrispBP [4]	-18.47	-8.92	8.60	9.52	18.47	11.62	0.23	0.52		
	airBP [26]	-19.58	-12.41	6.74	9.80	19.58	14.02	0.27	0.38		
	BP ³ (ours)	-3.09	-0.06	9.39	8.14	7.73	6.47	0.84	0.74		
Hypotension	CrispBP [4]	23	17.26	5.92	6.79	23	17.26	0.31	-0.07		
	airBP [26]	14.76	10.88	6.23	11.37	14.76	12.73	-0.20	0.74		
	BP ³ (ours)	-3.08	-0.91	8.54	7.49	7.85	5.90	0.86	0.75		

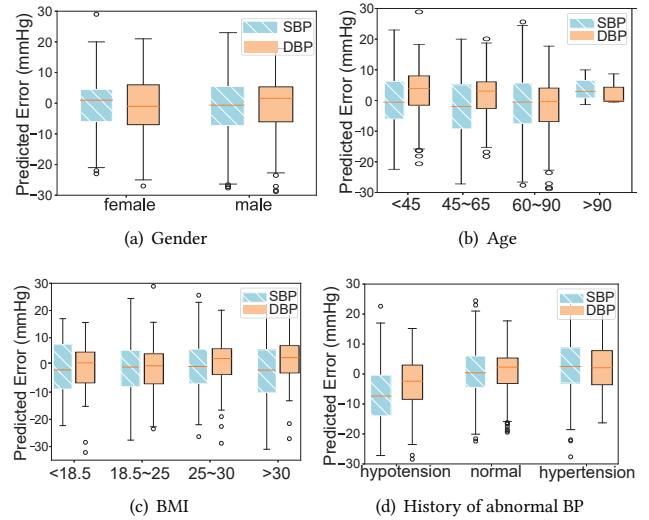
Table 3: Our performance on different dataset.

Dataset	Metrics (mmHg)		ME (SBP DBP)		STD↓ (SBP DBP)		MAE↓ (SBP DBP)		PCC↑ (SBP DBP)	
Laboratory	0.40	0.91	6.16	5.67	4.55	4.55	0.91	0.92		
Hospital	-0.29	-0.99	9.96	8.03	7.82	6.47	0.87	0.67		
Overall	-1.57	-0.34	9.77	7.93	7.74	5.69	0.82	0.82		

methods have good overall performance numerically at first glance, they are unable to distinguish between high and low levels of BP. Figure 13 demonstrates the performance of airBP as a representative. We can observe that the results are centered around three mean values, and the prediction range is quite narrow. Specifically, SBP predictions are centered around 100, 125, and 157 mmHg, which correspond to the average BP values of groups with hypotension history, normal BP, and hypertension history, respectively. This phenomenon, supported by the previous measurement results in Figure 4, indicates that existing models fail to learn meaningful information related to BP despite the inclusion of physiological factors.

Average accuracy over patients. We then conduct separate tests on subjects with Grade 2 hypertension (SBP > 160 or DBP > 100 mmHg) and hypotension (SBP < 90 and DBP < 60 mmHg), which represented 5.73% and 3.81% in our dataset respectively. As illustrated in the second and third major rows in Table 2, BP³ maintains almost consistent accuracy with overall performance, but other methods degrade significantly. Quantitatively, BP³ can accurately track BP levels (e.g. the ME of SBP and DBP of ours is 16.73% and 0.67% of Crisp-BP, 15.78% and 0.48% of airBP on Grade 2 hypertension groups) and have high reliability (e.g. the MAE of SBP and DBP of ours is 41.85% and 55.68% of Crisp-BP, 39.48% and 46.15% of airBP). *The results highlight the reliability and effectiveness of BP³, especially for scenarios involving subjects with abnormal BP who have a more urgent need for BP monitoring.*

Accuracy across different data. Our dataset comprises two distinct sections: laboratory data and hospital data. To assess its effectiveness across both settings, we conduct separate tests on each part of dataset. Additionally, we also perform overall experiments using a combined dataset that merges both sections of data.

**Figure 14: Predicted error across various physiological factor classifications.**

As presented in Table 3, under the laboratory dataset of 82 subjects, our performance meets the FDA's AAMI standard, with ME and STD are 0.40 and 6.16 mmHg for SBP, 0.91 and 5.67 mmHg for DBP. When applied to the hospital dataset, which include 930 patients, we observe a slight reduction in performance, which can be attributed to the more complex physiological factors and broader range in BP distribution. Moreover, our overall experiment surpasses the results achieved solely in hospital settings, illustrating that the amalgamation of diverse data sources can enhance the model's learning capacity.

Accuracy across diverse physiological factors. Figure 14 presents box-and-whisker plots illustrating the predicted BP values across various physiological factor categories. The box itself spans from the first quartile (Q1) to the third quartile (Q3), with a horizontal line indicating the median (Q2). The ends of the box plot extend to the minimum and maximum values and data points outside this range are considered outliers. We can observe that:

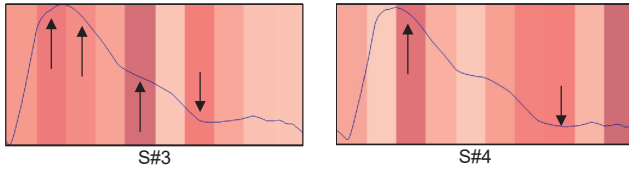


Figure 15: Visualization of attention mechanism on subjects with similar waveform.

(i) Our methodology demonstrates stable performance across diverse physiological states, with a MAE of less than 8 mmHg across all tested groups. (ii) Interestingly, the error margin for subjects at the extreme end of the age spectrum is remarkably small, with all errors not exceeding 10 mmHg. Despite these subjects constituting a smaller proportion of our dataset, this suggests our approach is effective across all age groups. (iii) For subjects with a history of BP issues, our method continues to deliver robust results with a slight decrease in accuracy. Specifically, the ME for these groups is -1.67 mmHg for SBP and -1.38 mmHg for DBP, the STD is 9.79 mmHg for SBP and 8.64 mmHg for DBP. (iv) Across all groups of subjects, the estimation for obese subjects poses the greatest challenge. This is primarily due to their radial artery tends to be situated deeper beneath the skin, and the minor vibrations that reach the skin surface are more likely to be obscured by noise.

5.2 Micro-benchmark Experiments

How BP³ solves multivalued mapping problem. By fusing the pulse waveform and personalized physiological factors, BP³ tackles the problem of multivalued mapping and enhances the accuracy of BP estimation. Here, we harness the visualization ability of attention mechanism to evaluate the effectiveness of our method in distinguishing between highly similar pulse waveforms based on physiological factors. Specifically, we upsample and normalize features extracted by the physio-pulse attention. As illustrated in Figure 15, we demonstrate the visual attention on pulse wave patches of S#3 and S#4, whose physiological information is shown in Figure 5. With darker colors indicating higher attention, we can observe that the model places emphasis on the upstroke, the early reflected wave and the position of dicrotic notch of S#3, who is an elderly male diagnosed with hypertension. In contrast, S#4 is a healthy female, resulting in the waveform focusing only on the highest and lowest patches in the waveform. The observation confirms that BP³ is effective in dealing with waveforms that are extremely similar but pay specific attention to different parts of the waveform due to physiological factors.

Impact of the patch size. We compare the impact of different patch sizes on system performance in Figure 16. In general, as the patch size p decreases, the MAE of SBP diminishes, indicating that smaller patches facilitate a more fine-grained representation of the input data. If $p < 16$, MAE remains low with MAE of DBP consistently under 6 mmHg, but further reductions yield only marginal improvements while each token must be processed individually, leading to a significantly higher computational load. Conversely, increasing patch size leads to performance degradation, as larger patches aggregate more data points and this coarser representation can obscure fine distinction in the pulse waveform. Thus, we select

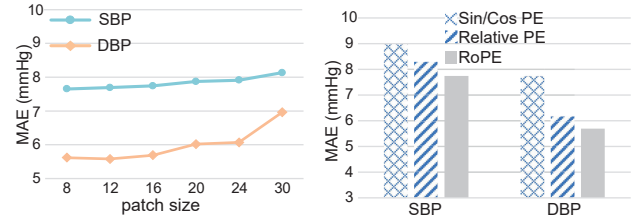
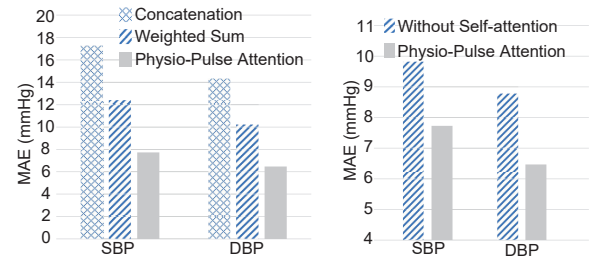


Figure 16: Impact of patch size. **Figure 17: Impact of position embedding choice.**



(a) Effectiveness of physio-pulse attention (b) Effectiveness of self-attention

Figure 18: Effectiveness of physio-pulse attention mechanism.

a patch size of $p = 16$ to achieve balance between the performance and computational efficiency.

Effectiveness of RoPE. To validate the effectiveness of the design of RoPE, we compare its performance against sin/cos position embedding (sin/cos PE) [52] and relative position embedding (relative PE) [45] in Figure 17. The superior performance of RoPE can be attributed to its ability to effectively preserve the relative positional information while capturing the periodicity of tokens. In contrast, the performance of sin/cos position embedding is found to be the least effective as it assigns absolute positions to tokens, limiting its capability in capturing the temporal dynamics of pulse waveforms. While relative position embedding offers advantages over sin/cos PE by considering token positions relative to each other, it can not model the inherent periodicity presented in pulse waveforms.

Effectiveness of physio-pulse attention. Few previous methods use concatenate [26] or weighted sum [4] to take into account personal information. However, they fundamentally neglect the critical relationship between the pulse waveforms and physiological factors. Hence, we design physio-pulse attention to leverage the guiding influence of the latter. In Figure 18(a), we conduct a comparative analysis of our method against both feature layer concatenation and weighted sum approaches. The results indicate that: (i) Feature layer concatenation exhibits the poorest performance in estimating BP in the Grade 2 hypertension group, with MAE of 17.29 mmHg for SBP and 12.39 mmHg for DBP, respectively. (ii) The weighted sum method shows a marginal improvement over feature layer concatenation, yet it still falls short of meeting the desired standards. For the Grade 2 hypertension group, the estimated SBP and DBP present MAE of 14.37 mmHg and 10.22 mmHg, respectively. The primary shortcoming of these two methods lies

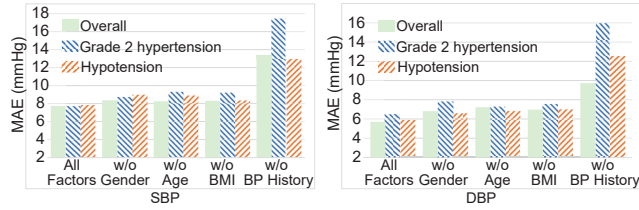


Figure 19: Impact of various physiological factors on MAE results.

in their inability to fully integrate waveform features with physiological factors, resulting in model confusion due to the multivalued mapping problem.

Effectiveness of self-attention. The physio-pulse attention module consists of two integral components: the cross-attention mechanism and the self-attention mechanism. The cross-attention mechanism is adept at learning how physiological factors influence different segments of the waveform. Conversely, the self-attention mechanism is crucial for identifying and establishing long-range dependencies across patches of the waveform. In scenarios where only the cross-attention mechanism is employed, there is a decline in performance as Figure 18(b) shows, with the MAE for SBP and DBP in subjects with Grade 2 hypertension, increasing to 9.82 mmHg and 8.78 mmHg, respectively. This demonstrates that if the model relies solely on the cross-attention mechanism, it will lack deeper insights into the internal structure of the waveform itself, leading to decreased performance. The results validate the complementary nature of the two mechanisms.

Impact of each physiological factor. To assess the influence of each physiological factor, we conduct tests by sequentially excluding one factor at a time: gender, age, BMI, and history of abnormal BP. We obtain results across various populations, including the overall dataset, subjects with Grade 2 hypertension, and subjects with hypotension as shown in Figure 19. The findings reveal that omitting any physiological factors results in diminished performance. Among these factors, a history of abnormal BP exerted the most significant influence on the outcomes, with the MAE of 13.31 mmHg, 17.65 mmHg, 12.95 mmHg for SBP and 9.73 mmHg, 16.09 mmHg, 12.56 mmHg for DBP across the three groups. This is primarily because a history of abnormal BP provides crucial context regarding the individual's structure and elasticity of blood vessels, both of which greatly impact BP readings. Age is the second most important factor, especially among subjects with Grade 2 hypertension, as hypertension is more prevalent in older individuals [32]. Without age as a factor, the MAE for the three groups is 8.24 mmHg, 9.37 mmHg, 8.98 mmHg for SBP and 7.22 mmHg, 7.28 mmHg, 6.87 mmHg for DBP.

6 Limitations and Future Works

Evaluation on dynamic activity states. The data-set collected in our work require subjects to stay in a static, relaxed state, with our method designed for nearly constant heart rates. Future research could prioritize evaluating the performance of the system in more dynamic contexts, such as before and after exercise or during various daily activities. This approach will help assess the

robustness and adaptability of our method over extended duration and in varying physiological conditions.

Evaluation of radar distance from the wrist The wearable prototype is designed to be placed close to the skin to enhance measurement sensitivity, facilitating better detection of pulse waves. We employ subsequent signal processing algorithms to improve the quality of detected signals. While near-filed sensing may present challenges [16], as signals may be influenced by reflections and scattering from the skin or surrounding tissues, complicating data interpretation. Future evaluations will further assess the device's performance at varying distances.

Impact of diseases on pulse waveforms. BP³ focused on four fundamental physiological factors: gender, age, BMI, and history of abnormal blood pressure. In addition, the influence of diverse diseases and related pharmacological treatments, particularly for patients suffering from cardiovascular disease, can significantly alter the characteristics of pulse waveforms. For example, heart failure might present a different pulse wave pattern compared to hypertension, beta-blockers or ACE inhibitors can affect heart rate, vascular resistance and so on, all of which are reflected in the characteristics of pulse waveforms. Our future work aims to unravel the complex relationship between pulse waveform and various diseases more comprehensively. By doing so, we aspire to enhance the precision of blood pressure monitoring across a broad spectrum of the population, including individuals suffering from related diseases.

Accuracy of BP ground-truth labels. An individual's physical condition changes over time, and BP values fluctuate with each heartbeat. Mapping a 35-second pulse wave to a single blood pressure value may compromise the accuracy of BP monitoring. Furthermore, while electronic cuff-based monitors offer a convenient solution for managing BP at home, their accuracy is limited in clinical settings. This is particularly hard for patients with arrhythmia, whose blood flow may present irregular status. As these devices may not provide reliable readings for patients and often fail to meet clinical gold standards, they may introduce noise into the labels used for developing predictive models. In future work, we will explore more reliable and continuous BP measurement devices such as cNIBP [15] or invasive methods in collaboration with medical professionals to provide more detailed and reliable ground-truth for blood pressure monitoring.

7 Related Work

mmWave Sensing. mmWave radar has numerous advantages in sensing, including a wide sensing range, high-resolution directional sensing capabilities, and resistance to adverse weather and lighting conditions [60]. Hence, various mmWave sensing technologies have been developed and applied to diverse human sensing applications in recent years, such as applications like human localization [6], gesture recognition [59], motion detection [25], and vital sign monitoring [57]. mmWave technology can detect movements ranging from centimeter-level respiratory motions to sub-millimeter arterial pulsations. For example, mmHRV [53] designs a contact-less multi-user heart rate variability monitoring system. Additionally, AirECG [61] leverages non-contact mmWave sensing to reconstruct electrocardiograms for both healthy individuals and cardiac patients with abnormal ECG waveforms, such as arrhythmia and

other heart conditions. mmBP [46] addresses the challenges of noise and minor movements that affect the mmWave wave radar signal quality and implements a contact-less BP monitoring system.

Pulse Wave Analysis. PWA methods, employing wireless technologies for the detection of pulse waves and extracting features related to BP, represent a rising area in cuff-less BP monitoring. These methods typically allow for measurements to be taken with the user being almost unaware of the process. Tian *et al.* [50] delve into the utilization of wave characteristics derived from BCG for blood pressure analysis. iPPG [8] extracts time domain, area, amplitude and derivatives features from facial video. While these methods show great potential, their practical utility and accuracy remains to be validated in a real-world, large population. Ran *et al.* [43] propose a dual-task network for pulse waveform reconstruction and BP estimation. The network uses an encoder to extract pulse features from radar signals and a decoder to reconstruct the waveform. Liu *et al.* [28] develop a CNN-Transformer hybrid architecture with a handcrafted feature-guided attention module, integrating manually extracted features and combining learned features with demographic data via a feature fusion module.

Multimodal Fusion. Recently, multimodal fusion has received widespread attention in many domains. Lu *et al.* [30] allow communication between frame-level and action-level information by cross-attention to predict frame-wise action labels of a video. Tang *et al.* [49] design a novel spatio-temporal criss-cross attention block for processing pose estimation in a video sequence. While there are numerous works on multimodal fusion, most of them are applicable to non-medical applications, such as combination of visual, textual, and audio data [58]. Cosmo [39] employs a two-stage training strategy for comparative fusion learning in multimodal human activity recognition, extracting both consistent and complementary information across different modalities. Wei *et al.* [56] propose a new multimodal cross-attention network for image and sentence matching, which can utilize the intra-modal relationship, and the inter modal relationship between image regions and sentences. Praveen *et al.* [40] propose the recursive joint cross-modal attention mechanism, which captures both intra-modal and inter-modal relationships across audio, visual, and text modalities. Our method is tailored for time-series pulse wave data and indicator personal physiological information, improving the performance and interpretability of blood pressure measurement.

8 Conclusion

From the dataset construction to the method development, we have built a BP measurement system named BP^3 in this paper, capable of predicting accurate readings robustly across diverse population groups. Utilizing commercial mmWave device, we have built the largest mmWave-BP dataset to our knowledge. This dataset encompasses pulse wave and physiological factors from thousands of individuals, the majority of whom are actual patients with cardiovascular disease. Through analysis of this large-scale dataset, we uncover the phenomenon of multivalued mapping between pulse waveforms and BP values, challenging the effectiveness of PWA methods. To address this, we explore the intricate relationship between physiological factors and pulse waveforms and design a customized multimodal frame, including different processing for temporal signals and indicator data, as well as a novel attention

mechanism that can guide the model to learn the subtle differences between waveforms based on different physiological factors. Evaluation on our dataset shows that the estimation of BP^3 is close to the FDA's AAMI standard. We observe significant improvements over state-of-the-art methods, especially for subjects with abnormal BP. Overall, it is the first time that the critical role of physiological factors in PWA methods is highlighted, underscoring their importance in enhancing the accuracy and reliability in BP estimation.

Acknowledgments

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